For processes like Mess3 that have non-zero row sums for each labeled transition matrix, this can be written more simply as:

$$\vec{r}_1^{(z_{1:\mathrm{d}})} = \boldsymbol{\pi} + \sum_{n=0}^{\mathrm{d}-1} \left(\boldsymbol{\pi} T^{|z_{\mathrm{d}-n}} T^n - \boldsymbol{\pi} \right) , \tag{27}$$

which is the form that appears in the main text. For other processes that don't satisfy this condition, slight modifications of the equations in the main text follow straightforwardly from Eq. (26).

B. Model architecture and training procedure

We employ a standard single-layer transformer model with learned positional embeddings. The model architecture follows the conventional transformer design, with $d_{\text{model}} = 64$ and $d_{\text{ff}} = 256$. Depending on the Mess3 parameters, we use either a single-head or a double-head attention mechanism. We conduct a systematic sweep over the HMM parameters α and x, training a separate model for each pair. Models are trained on next-token prediction using cross-entropy loss, with batch size 128. We use Adam optimizer (Kingma & Ba, 2017) with a 10^{-4} learning rate and no weight decay. Each model is trained for approximately 15 million tokens.

We generate all possible input sequences up to length 10, recording hidden activations from the transformer's residual stream. These activations are organized into a dataset capturing the model's response to all input patterns.

Input sequences consist of three symbols, embedded with positional information, without a beginning-of-sequence (BOS) token.

C. Recurrent vs. Parallel Belief Updating

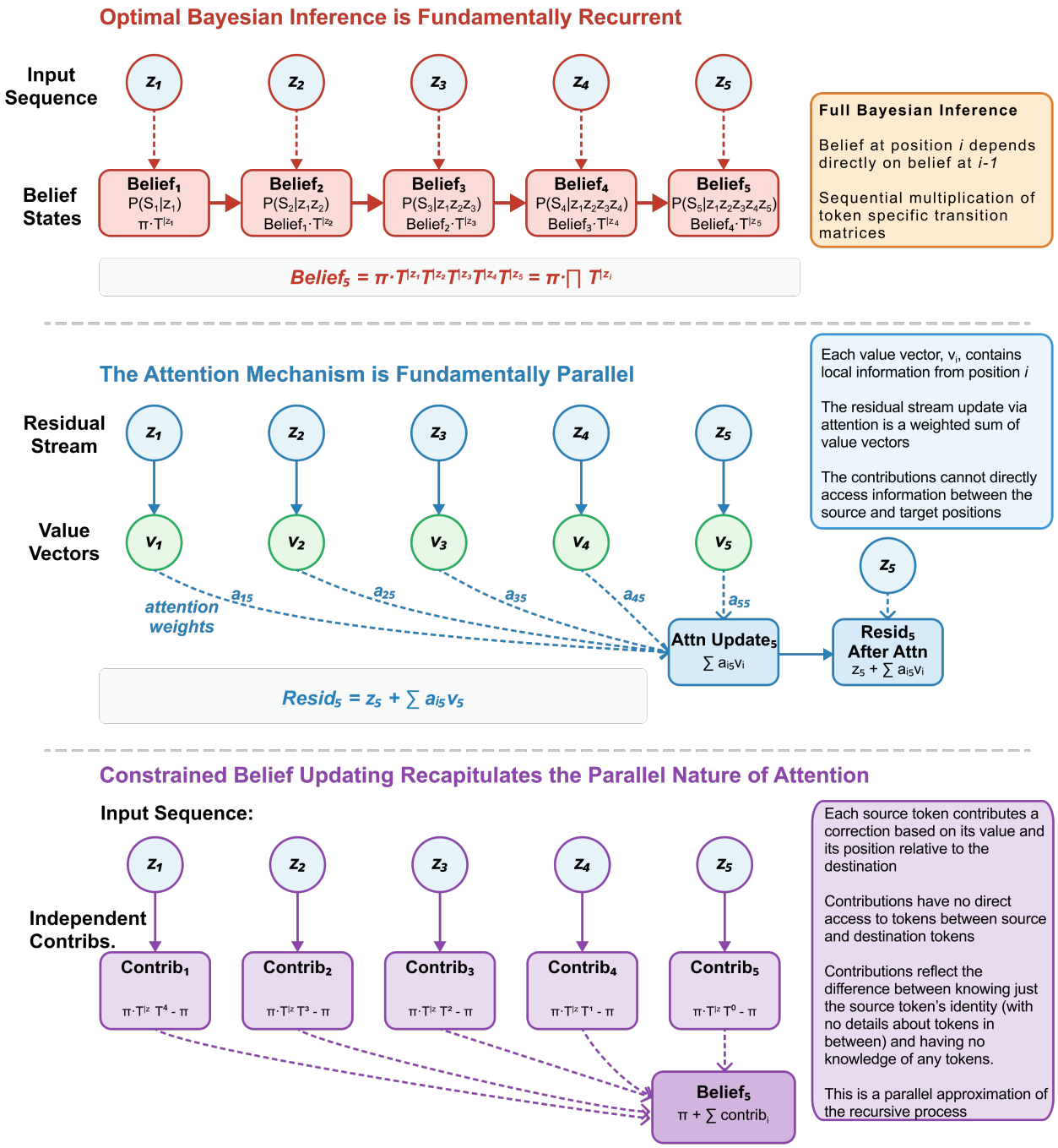

Figure A1: Diagrammatic visualization comparing the recurrent nature of Optimal Bayesian Inference (top), to the parallel attention mechanism (middle), and the parallel Constrained Belief Updating presented in this paper.

D. Quantification of Theoretical Predictions

See Figs. A2, A3, and A4.

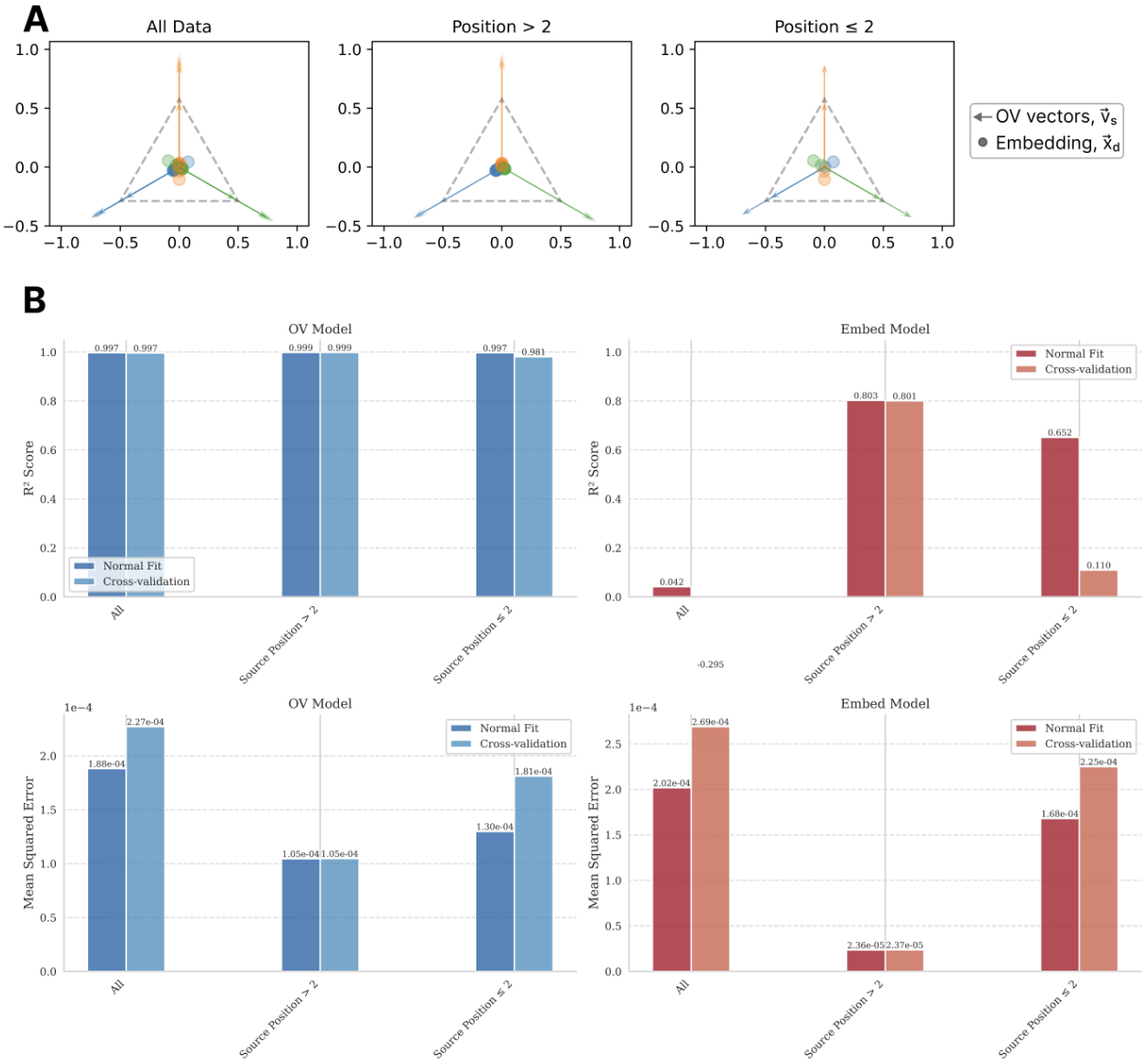

Figure A2: Embeddings for the first two positions are correctly predicted to be parallel to the OV vectors, as with all of the embeddings; however the sign of the predicted embedding for these first two positions deviates from the observed embedding. We do not yet understand the reason for this discrepancy, but still find it remarkable that the bulk of the high-dimensional computation carried out by attention—attention pattern, OV vectors, and all embeddings beyond the first two positions—can be very precisely understood by a sequence of operations in the two-dimensional simplex.

E. Multi-layer experiments

See Figs. A5 and A6.